

E028

Hybrid Cloud Infrastructure and Orchestration

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A manufacturer runs its ERP in its own data center and connects it to analytics services in one public cloud, managed together with shared policies. The IT director has 20 years of experience and a new office. Which term best describes this environment?

- A. Private cloud.
- B. Hybrid cloud.
- C. Community cloud.
- D. Multi-cloud.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A startup with no data center runs its app in one public cloud and its data warehouse in a second public cloud. Which term fits?

- A. Multi-cloud.
- B. Hybrid, since any two environments count.
- C. Hybrid cloud.
- D. Edge computing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A video studio stores 2 PB of raw footage on-premises and edits it daily with a local render farm. A consultant suggests moving the editing app to public cloud while leaving the footage in place. The studio won three awards last year. What is the main problem with the suggestion?

- A. Move the footage to cloud and keep editing on-premises.
- B. The awards show the current setup should never change.
- C. Data gravity: the app would be far from 2 PB of data it uses constantly, causing latency and transfer cost.
- D. Public cloud cannot run video software.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A team manages its hybrid network with declarative IaC in Git. During an outage, an engineer changed a firewall rule by hand in the cloud console. The engineer works from the Denver office. What will likely happen on the next IaC run?

- A. Nothing, because the change came from Denver.
- B. The tool adopts the manual change automatically.
- C. The run fails permanently.
- D. The tool detects drift and reverts the rule to what the code declares unless the code is updated.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

An engineer runs the same declarative IaC configuration twice in a row with no code changes. What should the second run do?

- A. Fail, since configurations can run only once.
- B. Delete everything and rebuild.
- C. Create a duplicate set of resources.
- D. Make no changes, because the environment already matches.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A retailer's release process provisions VMs on-premises, deploys containers to public cloud, updates a load balancer, and runs tests in order, rolling back on failure. The process runs on a server named ORCH-01. What is this?

- A. Automation of a single task.
- B. Orchestration.
- C. Orchestration, because of the server's name.
- D. Manual operations.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A script automatically rotates one server's log files every night. Which term fits best?

- A.** Orchestration.
- B.** Infrastructure as Code.
- C.** Automation.
- D.** Cloud bursting.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A hospital must copy 40 TB of imaging archives to a cloud region over a 1 Gbps link that averages 80% efficiency. The archive sits on a storage array bought in 2019. About how long will the transfer take?

- A.** About 13.9 hours
- B.** About 4.6 days
- C.** About 3.7 days
- D.** Overnight, since any modern link moves 40 TB in hours.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A team must move 18 TB within 48 hours over a link that runs at 80% efficiency. What is the minimum link speed?

- A.** About 130 Mbps
- B.** About 667 Mbps
- C.** About 833 Mbps
- D.** About 1.04 Gbps

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An insurer runs apps on-premises and in two public clouds. Each environment has its own user directory and monitoring console, and a former employee still had access in one cloud a month after leaving. The insurer's headquarters is being renovated. What is the best fix?

- A.** Federate identity from one directory to all environments and centralize logging and monitoring.
- B.** Delay changes until the renovation is done.
- C.** Add a fourth directory for contractors.
- D.** Audit the one cloud where the account remained.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A tax-preparation company's on-premises servers handle load most of the year, but traffic triples during the six weeks before the filing deadline. Its main app is containerized and already deploys to a public cloud test account. The company's CFO is a CPA. What strategy fits?

- A.** Buy three times the on-premises servers.
- B.** Move everything to edge devices.
- C.** Cloud bursting: overflow to public cloud during the peak weeks.
- D.** Do nothing, since peaks always resolve themselves.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An oil company has sensors on 200 remote wells producing gigabytes per hour. A valve must shut within 50 milliseconds of a pressure spike. The nearest cloud region is 1,200 miles away, and the company's logo is orange. Where should the shutdown decision run?

- A.** On edge devices at each well, sending summaries to the cloud.
- B.** Send all raw data to the cloud, since bandwidth is unlimited.
- C.** In the head-office data center.
- D.** In the distant cloud region.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A manufacturer runs its ERP in its own data center and connects it to analytics services in one public cloud, managed together with shared policies. ~~The IT director has 20 years of experience and a new office.~~ Which term best describes this environment?

Target word: classify the environment **Decides:** On-premises plus public cloud, integrated.

A	Dropped element	Drops the public cloud part.
B	Correct	On-premises integrated with public cloud.
C	Adjacent category	Shared by organizations with common needs.
D	Confusion set	Multi-cloud means more than one public provider.

2. Answer A Concept 1 • transfer

A startup with no data center runs its app in one public cloud and its data warehouse in a second public cloud. Which term fits?

Target word: reverse: two public clouds, no on-premises **Decides:** No private or on-premises component.

A	Correct	Two public providers.
B	Absolute language	The environments must include private or on-premises.
C	Inverted operation	Hybrid needs an on-premises or private component.
D	Adjacent category	About processing location near data sources.

3. Answer C Concept 2

A video studio stores 2 PB of raw footage on-premises and edits it daily with a local render farm. A consultant suggests moving the editing app to public cloud while leaving the footage in place. ~~The studio won three awards last year.~~ What is the main problem with the suggestion?

Target word: placement constraint **Decides:** Data gravity: the app needs fast access to huge local data.

A	Inverted operation	Same separation in reverse.
B	Camouflage	Awards do not decide placement.
C	Correct	Compute should sit near heavy data.
D	Absolute language	It can; placement is the issue.

4. Answer D Concept 3

A team manages its hybrid network with declarative IaC in Git. During an outage, an engineer changed a firewall rule by hand in the cloud console. ~~The engineer works from the Denver office.~~ What will likely happen on the next IaC run?

Target word: IaC behavior with drift **Decides:** Declarative IaC reconciles to the code.

A	Camouflage	Location is irrelevant.
B	Inverted operation	Declarative tools apply the code, not the console.
C	Absolute language	Drift is detected and corrected, not fatal.
D	Correct	Code is the source of truth.

5. Answer D Concept 3 · transfer

An engineer runs the same declarative IaC configuration twice in a row with no code changes. What should the second run do?

Target word: idempotency **Decides:** Desired state already met.

A	Absolute language	They are built to run repeatedly.
B	Adjacent category	Not required when state matches.
C	Inverted operation	That is non-idempotent scripting.
D	Correct	Idempotent runs produce the same result.

6. Answer B Concept 4

A retailer's release process provisions VMs on-premises, deploys containers to public cloud, updates a load balancer, and runs tests in order, rolling back on failure. ~~The process runs on a server named ORCH-01.~~ What is this?

Target word: classify the process **Decides:** Many coordinated tasks across systems.

A	Confusion set	Automation alone covers one task.
B	Correct	Coordinates multiple automated tasks into one workflow.
C	Camouflage	A name decides nothing.
D	Inverted operation	The steps are automated.

7. Answer C Concept 4 · transfer

A script automatically rotates one server's log files every night. Which term fits best?

Target word: reverse: single task **Decides:** One task, one system.

A	Inverted operation	Needs coordination across many tasks.
B	Confusion set	Defines infrastructure; log rotation is an operational task.
C	Correct	A single task without manual steps.
D	Adjacent category	About overflow capacity.

8. Answer B Concept 5

A hospital must copy 40 TB of imaging archives to a cloud region over a 1 Gbps link that averages 80% efficiency. ~~The archive sits on a storage array bought in 2019.~~ About how long will the transfer take?

Target word: transfer time **Decides:** $40 \times 10^{12} \times 8 \text{ bits} \div 0.8 \times 10^9 \text{ bits/s} = 400,000 \text{ s}$.

A	Confusion set	Treats bytes as bits (skips $\times 8$).
B	Correct	$400,000 \text{ s} \approx 111 \text{ hours} \approx 4.6 \text{ days}$.
C	Dropped element	Ignores the 80% efficiency.
D	Absolute language	The math says days.

9. Answer D Concept 5 · transfer

A team must move 18 TB within 48 hours over a link that runs at 80% efficiency. What is the minimum link speed?

Target word: solve for bandwidth **Decides:** $1.44 \times 10^{14} \text{ bits} \div 172,800 \text{ s} = 833 \text{ Mbps effective} \div 0.8$.

A	Confusion set	Forgets bytes-to-bits ($\times 8$).
B	Inverted operation	Multiplies by 0.8 instead of dividing.
C	Dropped element	Leaves out the efficiency loss.
D	Correct	$833.3 \text{ Mbps} \div 0.8 \approx 1,042 \text{ Mbps}$.

10. Answer A Concept 6

An insurer runs apps on-premises and in two public clouds. Each environment has its own user directory and monitoring console, and a former employee still had access in one cloud a month after leaving. ~~The insurer's headquarters is being renovated.~~ What is the best fix?

Target word: hybrid management fix **Decides:** Separate identity stores caused missed deprovisioning.

A	Correct	One identity source and one view close the gaps.
B	Camouflage	The renovation does not decide.
C	Inverted operation	Adds more silos.
D	Dropped element	Fixes one symptom, not the cause.

11. Answer C Concept 7

A tax-preparation company's on-premises servers handle load most of the year, but traffic triples during the six weeks before the filing deadline. Its main app is containerized and already deploys to a public cloud test account. ~~The company's CFO is a CPA.~~ What strategy fits?

Target word: strategy for short peaks **Decides:** Short peaks plus portable workload.

A	Inverted operation	Idle most of the year.
B	Adjacent category	Edge addresses latency near data, not seasonal load.
C	Correct	Pays for extra capacity only when needed.
D	Absolute language	Unhandled peaks mean outages.

12. Answer A Concept 8

An oil company has sensors on 200 remote wells producing gigabytes per hour. A valve must shut within 50 milliseconds of a pressure spike. The nearest cloud region is 1,200 miles away, ~~and the company's logo is orange.~~ Where should the shutdown decision run?

Target word: placement for real-time control **Decides:** 50 ms limit and heavy data at remote sites.

A	Correct	Local processing meets latency and cuts bandwidth.
B	Absolute language	Remote links are limited and costly.
C	Adjacent category	Still far from the wells.
D	Inverted operation	Round-trip delay risks missing the 50 ms limit.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Hybrid vs. multi-cloud	Hybrid cloud integrates on-premises or private cloud with public cloud as one managed environment; multi-cloud uses more than one public cloud provider. An organization can be both.	Calling any use of two public clouds “hybrid.”	Whether on-premises/private infrastructure is part of the mix.
2	Workload placement	Place each workload by its constraints: data gravity, latency, compliance, cost pattern, and dependencies — not by a blanket rule.	Moving everything to public cloud (or keeping everything on-premises) by policy.	A workload with a stated latency, data-volume, or regulatory constraint.
3	Infrastructure as Code	Declarative IaC describes the desired end state; the tool computes changes, runs idempotently, and lives in version control so environments are repeatable and drift is detectable.	Making manual console changes to IaC-managed resources.	An environment that differs from its code, or a script run twice.
4	Automation vs. orchestration	Automation performs a single task without manual steps; orchestration coordinates many automated tasks across systems into an ordered workflow, often reconciling to a desired state (e.g., Kubernetes).	Using the terms interchangeably.	One task vs. a multi-step, multi-system workflow.
5	Data transfer time	$\text{Time} = \text{data size in bits} \div \text{effective link rate (link speed} \times \text{efficiency)}$; 1 byte = 8 bits.	Forgetting to convert bytes to bits or ignoring link efficiency.	A data volume in TB and a link speed in Mbps or Gbps.
6	Unified identity and observability	A hybrid estate needs one federated identity system and centralized monitoring and logging across all environments.	Keeping separate user directories and consoles for each environment.	Separate accounts or blind spots across environments.
7	Cloud bursting	Bursting runs baseline load on-premises and overflows to public cloud at peaks; it requires portable workloads and adequate connectivity.	Assuming any on-premises app can burst without changes.	Short seasonal peaks above on-premises capacity.
8	Edge computing	Edge processes data near where it is created to cut latency and bandwidth, sending summaries to the cloud.	Sending all raw sensor data to a distant cloud region for real-time decisions.	Millisecond decisions or huge raw data volumes at remote sites.

Course scope (public description): Hybrid Cloud Infrastructure and Orchestration delves into the design, implementation, and management of hybrid cloud environments. It covers the integration of on-premises infrastructure, private cloud, public cloud, and edge computing to create a hybrid cloud ecosystem.